

- **To pass:** submit a serious attempt to answer the question.
 - **Hand in:** upload a zip-file with a PDF and code to the course webpage.
 - **Collaboration:** Work individually, but discussions are encouraged
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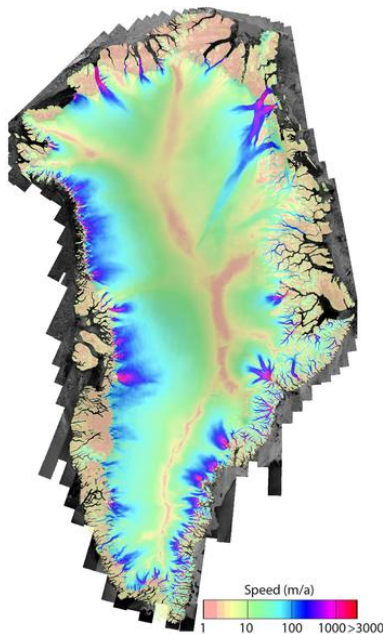


Figure 1: Surface velocities on the Greenland Ice Sheet constructed from 20 years of satellite observations, see doi.org/10.1017/jog.2017.73

Your task is construct an inverse model which determines the bedrock shape underneath an 800 meter thick ice sheet by using observations of surface velocities and surface height that you find on the course webpage. You can assume that the dynamics of the ice sheet is well described by the Shallow Ice Approximation.

Specifically, you should retrieve the amplitude of the two bumps in the bedrock. Assume the bedrock is created as in `icemodel.CreateBedrock`, and that the only uncertain parameters are the amplitudes of the bumps You steal any code you like from the forward model provided on the course webpage.

You should also investigate how measurement errors impacts your inverse model. Add some random noise (a few percent) to the data - how does that impact your ability to determine the bedrock amplitudes?

Hints:

- To test your model, you can create your own data with the forward model (uncomment the lines for saving data).
- Note that you do not know at what time-point the data was retrieved, i.e. you cannot use the transient free surface equation in your inverse model. This fact is actually making the task easier and more computationally efficient.

Your hand in should include:

- Your answer, i.e. the bedrock amplitudes
- A plot or table which explains why you chose those amplitudes
- A plot with the forward model for your chosen bedrock (run it for at least 200 years)
- A brief explanation of your method
- Your code
- Some plot or table which describes how noise impacts your inverse model, together with a few sentences.